

Mobile processor for gamers

Due to the availability of high-performing mobile systems, gaming enthusiasts are increasingly moving towards these



devices. Rich processing features offer an extraordinary gaming experience, something gamers greatly prefer nowadays.

Intel's new 10th Gen Intel Core processors that are loaded with an incredible performance capability, deliver faster performance for immersive gaming experience and amazing responsiveness. As games and applications continue to depend on high-frequency cores, through this launch, Intel has pushed the frequency boundary to achieve lower latency and deliver best PC gaming experience on a laptop.

Led by the 10th Gen Intel Core i9-10980HK processor, the H-series delivers a high-end performance. The processor features an excellent performance with up to 5.3GHz Turbo, eight cores, 16 threads and 16MB of Intel Smart Cache. The processor also allows further customisation, optimisation and tuning of a CPU's performance.

Intel
www.intel.in

POWER SUPPLIES

Gate driver IC

In Switching Mode Power Supplies (SMPS), ground potential is the reference for input signals of the gate driver IC. During switching of power MOSFETs,



parasitic inductances produce ground shifts that may cause false triggering of the gate driver integrated circuit (IC). Infineon Technologies AG adds a new IC to its cost-effective and compact size EiceDRIVER 1EDN TDI (truly differential inputs) 1-channel gate driver family to prevent such consequences. The latest gate driver IC 1EDN7550, launched by

the company, offers significant board-area savings, which together with its versatility makes it a very attractive part.

The EiceDRIVER 1EDN TDI can withstand static ground-shifts of up to $\pm 70V$ and transient ground shifts of up to $\pm 150V_{peak}$ with 3.3V PWM input signal at the application level. The EiceDRIVER 1EDN7550U enables 25V and 40V OptiMOS MOSFETs to operate in switched capacitor topologies at 1.2MHz switching frequency. A high power density of 3060W/in³ and a 97.1% peak efficiency (including auxiliary losses) have proven to be possible even in such applications.

The EiceDRIVER 1EDN TDI family is available in a standard SOT-23 6-pin package and the new device (1EDN7550U) is housed in an ultrasmall (1.5mm $\times$ 1.1mm $\times$ 0.39mm) 6-pin leadless TSNP package.

Infineon Technologies
www.infineon.com

Power management IC

Renesas Electronics Corporation has launched new Power Management IC (PMIC) reference designs to power the



multiple supply rails of Xilinx Artix-7 FPGAs, Spartan-7 FPGAs and Zynq-7000 SoCs, with and without DDR memory. The reference designs speed the development of power supplies for a variety of industrial and computing applications, including motor control, machine vision cameras, programmable logic controllers (PLCs), home gateways and appliances, and portable medical and wireless equipment.

Renesas' high-efficiency PMIC reference designs provide user-friendly solutions that support different Xilinx speed grades and DDR memory types: DDR3, DDR3L, DDR4, LPDDR2 and LPDDR3. The PMICs come in 4.7mm $\times$ 6.3mm, 35-ball BGA with 0.8mm pitch packages.

Renesas
www.renesas.com

30W power supplies

MORNSUN, a manufacturer of power supplies, has released open-frame, telecom power supplies VCB_SO-



3/6WR3 and VCB_SBO-10/30WR3. The series is high in performance, has a compact size and

high power density, and is equipped with a self-developed IC.

Due to its wide operating temperature range from $-40^{\circ}C$ to $+85^{\circ}C$, the series has an excellent temperature derating curve. The maximum working temperature is $85^{\circ}C$ with a seventy per cent load. It also features 1500V DC isolation voltage, 3mA no-load current and ninety per cent output efficiency.

It is widely used in the telecommunication field, such as FSU, switch, battery online monitoring, intelligent communication gateway, GPS synchronous clock, and 4G/5G base station.

Mornsun
www.mornsun-power.com

TEST & MEASUREMENT

5G NR test equipment

Before the 5G systems can be deployed, companies must test and validate products and services for standards compliance, interoperability, and operational performance. Hence, after the



initial non-standalone (NSA) 5G NR launch last year, National Instruments now offers

an upgraded stand-alone (SA) version of its 5G New Radio (NR) Test User Equipment (UE).

It operates as a 3GPP Release 15 standard compliant UE such as a modem or cell phone, allowing infrastructure providers to test, validate, and assess performance in real-world scenarios.

National Instruments
www.ni.com